

• Waves and Oscillation

• Wave:

Carrier of energy. (medium)

• Oscillation:

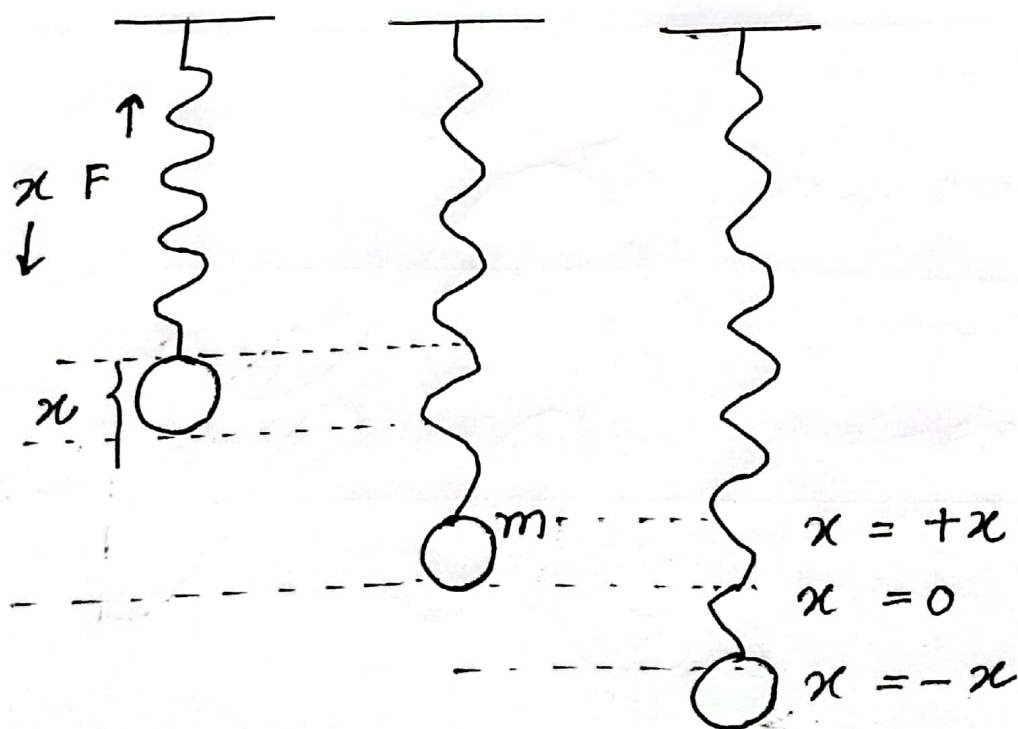
• Simple Harmonic Motion:

• Periodic Motion:

• Oscillatory Motion: One kind of periodic motion cross equilibrium ^{position} once or more.

• Simple Harmonic motion: Graphs

• mass-spring system:



From Hook's law, $[K = \text{spring constant}]$

$$F \propto -x$$

$$\Rightarrow F = -Kx \dots \dots \dots (i)$$

From Newton's 2nd Law,

$$F = ma \dots \dots \dots (ii)$$

From eqn. (i) and (ii),

$$ma = -Kx$$

$$\Rightarrow m \cdot \frac{dv}{dt} = -Kx$$

$$\left[a = \frac{dv}{dt} \right], \omega = v$$

$$\Rightarrow m \cdot \frac{d}{dt} \left(\frac{dx}{dt} \right) = -Kx \quad \left[v = \frac{dx}{dt} \right]$$

$$\Rightarrow m \cdot \frac{d^2x}{dt^2} = -Kx$$

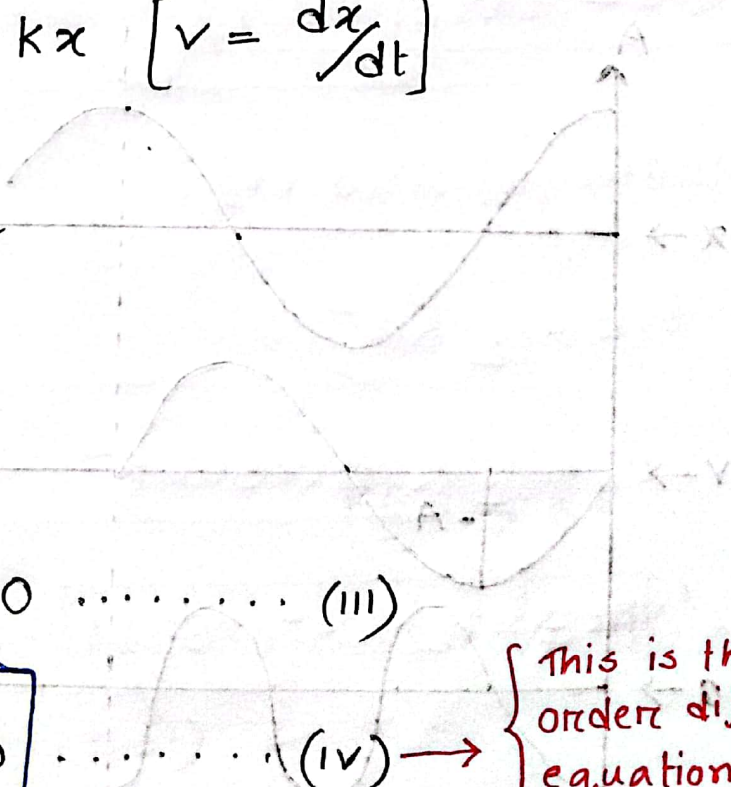
$$\Rightarrow \frac{d^2x}{dt^2} = -\frac{K}{m} x$$

$$\Rightarrow \frac{d^2x}{dt^2} + \frac{K}{m} x = 0 \dots \dots \dots (iii)$$

$$\Rightarrow \frac{d^2x}{dt^2} + \omega_0^2 x = 0 \dots \dots \dots (iv)$$

$\omega_0 = \sqrt{\frac{K}{m}}$ $\omega_0 = \text{Angular frequency}$

This is the 2nd order differential equation of motion



Solution of equation (iv) $[A = \text{max displacement}]$

$$\boxed{x = A \cos(\omega_0 t)} \dots \dots \dots (v)$$

$$v = \frac{dx}{dt} = \frac{d}{dt} \{A \cos(\omega_0 t)\}$$

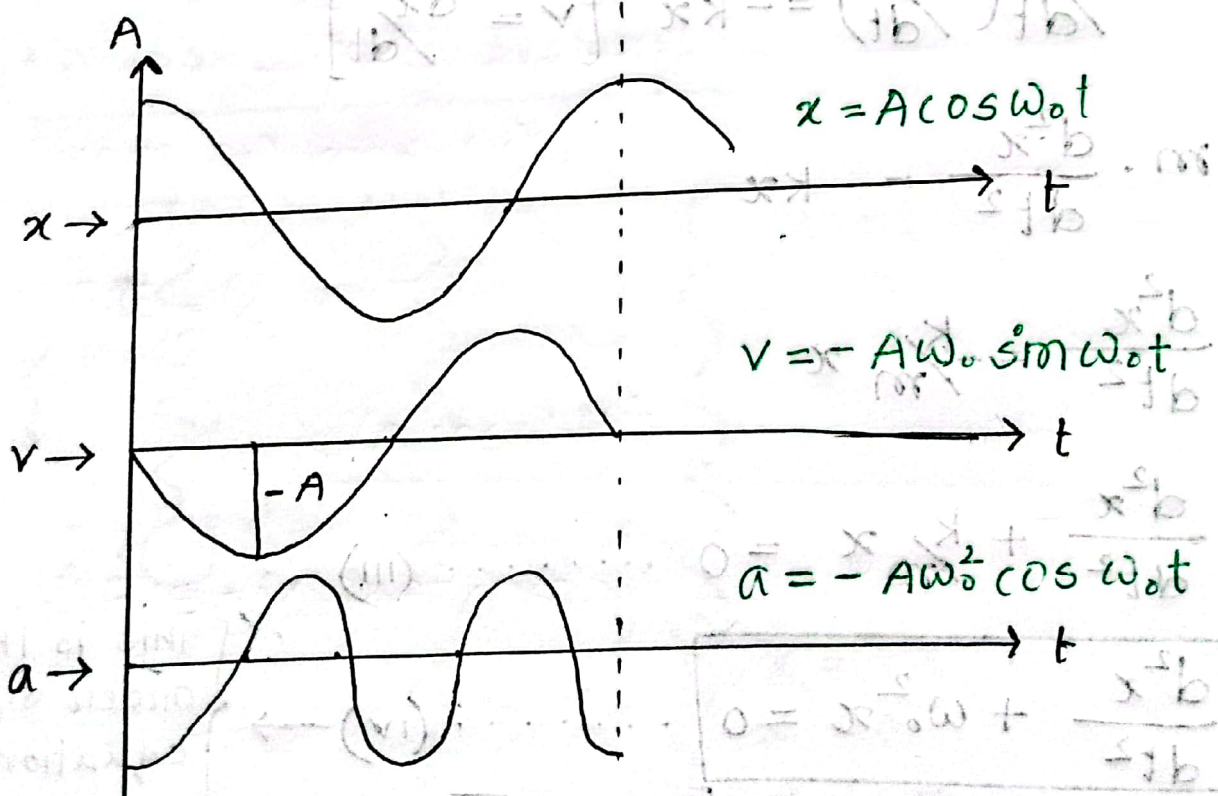
$$\boxed{v = -A\omega_0 \sin(\omega_0 t)} \dots \dots \dots (vi)$$

$$\therefore a = \frac{dv}{dt} = \frac{d}{dt} \{-A\omega_0 \sin(\omega_0 t)\}$$

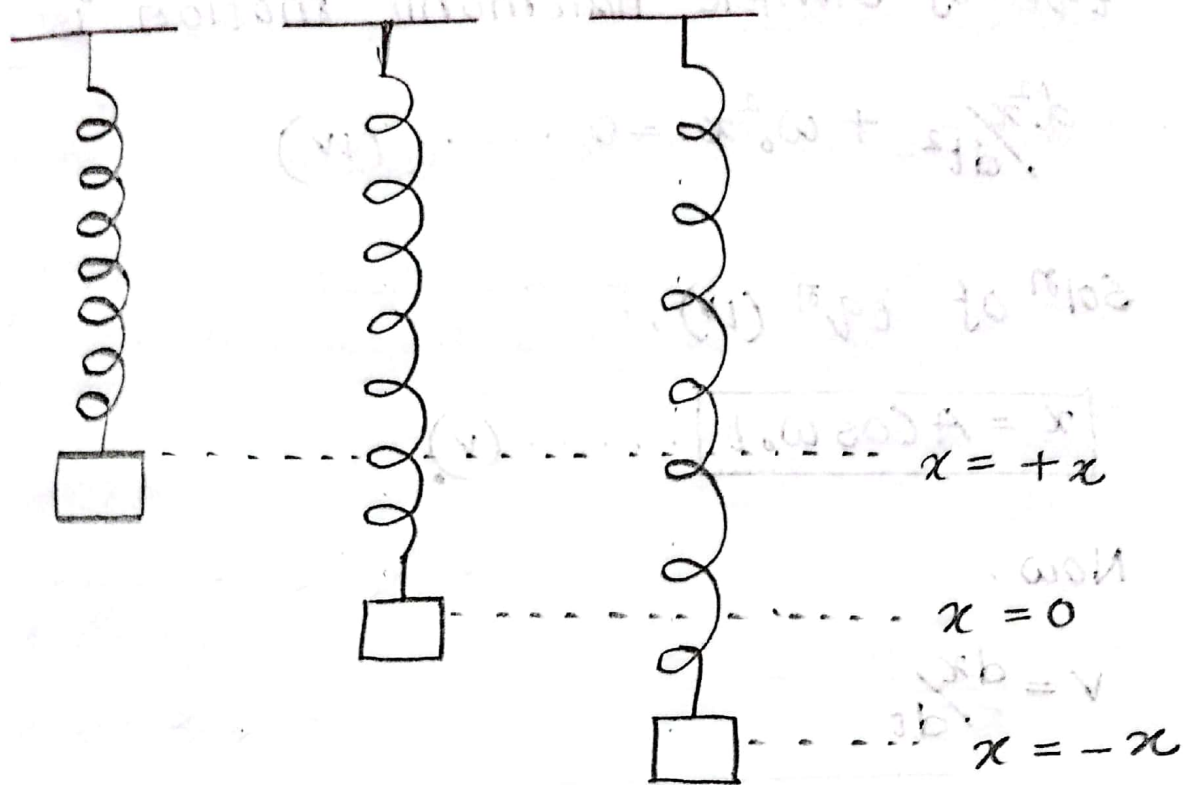
$$\therefore \boxed{a = -A\omega_0^2 \cos(\omega_0 t)} \dots \dots \dots (vii)$$

$$\boxed{v = \omega_0 \sqrt{A^2 - x^2}}$$

Graphs



Energy for mass-spring system.



$$P.E = \int_0^{+x} F \cdot dx$$

$$= \int_0^{+x} -kx \cdot dx$$

$$= -k \left[\frac{x^2}{2} \right]_0^{+x}$$

$$= -k \left[\frac{x^2}{2} - \frac{0^2}{2} \right]$$

$$= -\frac{1}{2} kx^2 \dots \dots (i)$$

$$P.E = \frac{1}{2} kx^2 \dots \dots (ii)$$

$$P.E = \frac{1}{2} mv^2 \dots \dots (iii)$$

We know from mass spring system,
eqn. of simple harmonic motion is,

$$\frac{d^2x}{dt^2} + \omega_0^2 x = 0 \dots\dots (iv)$$

Soln of eqn (iv),

$$\boxed{x = A \cos \omega_0 t} \dots\dots (v)$$

Now,

$$v = \frac{dx}{dt}$$

$$= \frac{d}{dt} (A \cos \omega_0 t)$$

$$\boxed{v = -A \omega_0 \sin \omega_0 t} \dots\dots (vi)$$

From (ii) and (v),

$$P.E = \frac{1}{2} kx^2$$

$$= \frac{1}{2} k (A \cos \omega_0 t)^2$$

$$\therefore \boxed{P.E = \frac{1}{2} k A^2 \cos^2 \omega_0 t} \dots\dots (vii)$$

From (iii) and (vi)

$$K.E = \frac{1}{2} m v^2$$

$$= \frac{1}{2} m (-A \omega_0 \sin \omega_0 t)^2$$

$$\therefore \boxed{K.E = \frac{1}{2} m A^2 \omega_0^2 \sin^2 \omega_0 t} \dots \dots (viii)$$

Total energy,

$$(vii) + (viii) \Rightarrow$$

$$E = P.E + K.E$$

$$= \frac{1}{2} K A^2 \cos^2 \omega_0 t + \frac{1}{2} m A^2 \omega_0^2 \sin^2 \omega_0 t$$

$$= \frac{1}{2} K A^2 \cos^2 \omega_0 t + \frac{1}{2} m A^2 (K/m) \sin^2 \omega_0 t$$

$$= \frac{1}{2} K A^2 (\cos^2 \omega_0 t + \sin^2 \omega_0 t)$$

$$= \frac{1}{2} K A^2$$

$$\therefore \boxed{E = \frac{1}{2} K A^2}$$

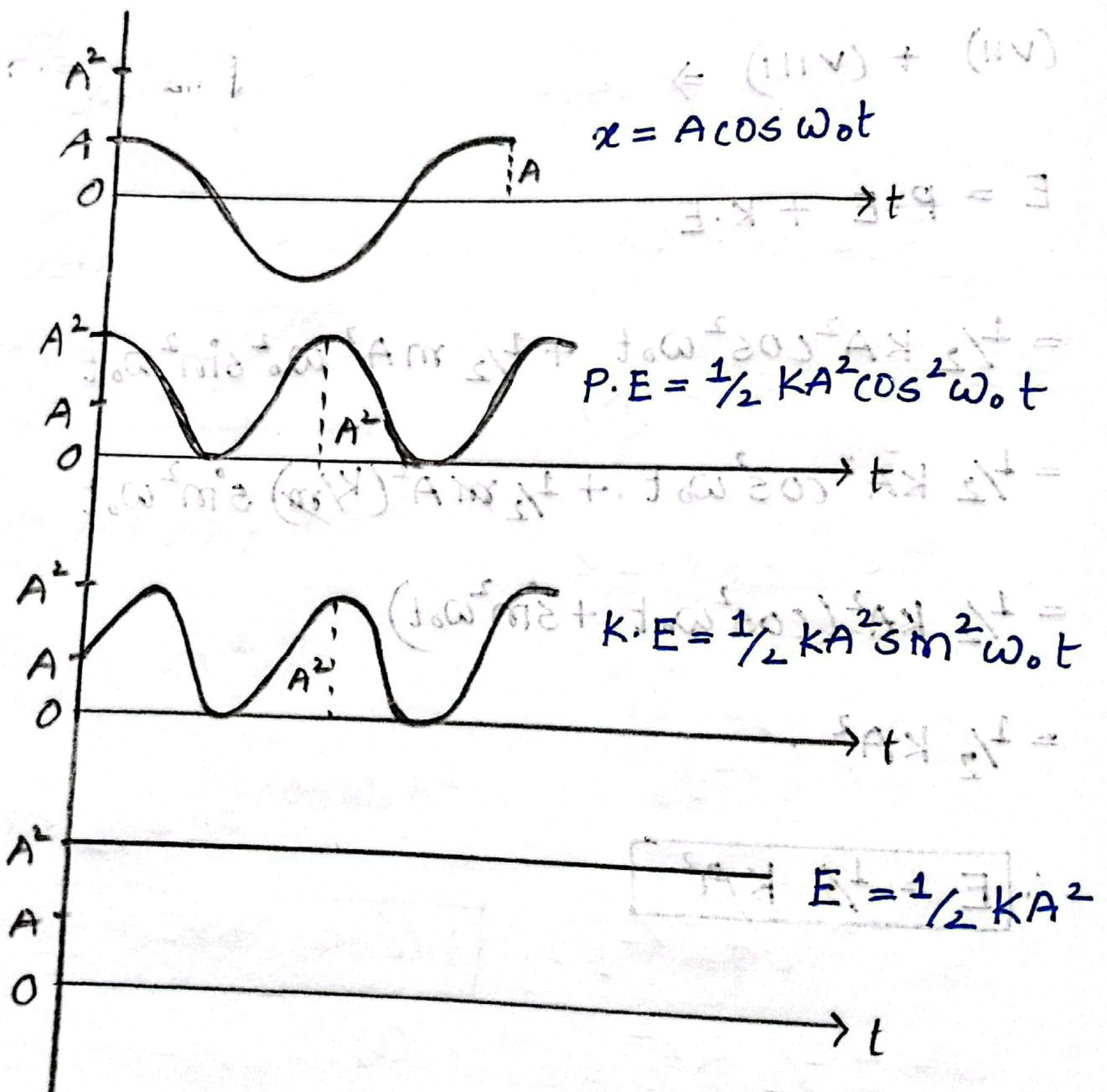
Graphs

$$x = A \cos \omega_0 t$$

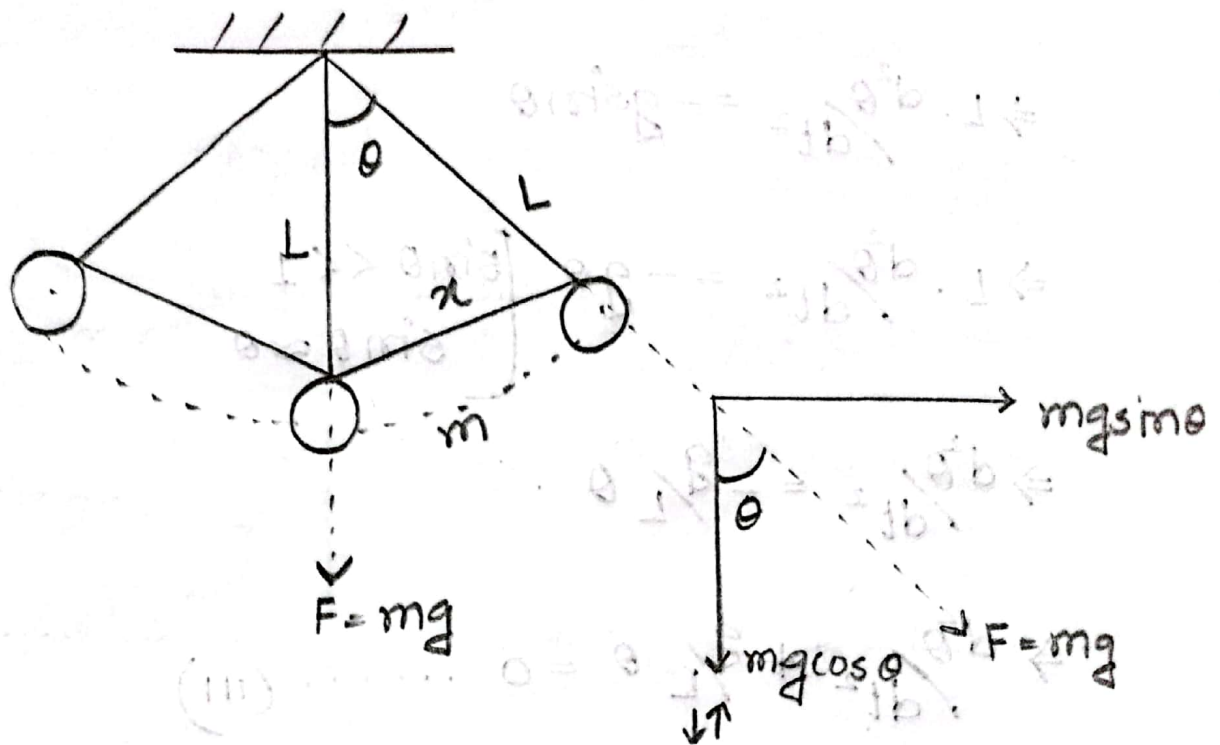
$$P.E = \frac{1}{2} K A^2 \cos^2 \omega_0 t$$

$$K.E = \frac{1}{2} K A^2 \sin^2 \omega_0 t$$

$$E = \frac{1}{2} K A^2$$



Simple Harmonic Motion: Simple Pendulum



$$F = -mg \sin \theta \dots \dots \dots (i)$$

From newton's 2nd law,

$$F = ma \dots \dots \dots (ii)$$

From eqn. (i) and (ii)

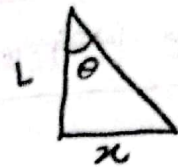
$$F = -mg \sin \theta$$

$$\Rightarrow ma = -mg \sin \theta$$

$$\Rightarrow m \frac{d^2x}{dt^2} = -mg \sin \theta \quad \left[a = \frac{d^2x}{dt^2} \right]$$

$$\Rightarrow \frac{d^2x}{dt^2} = -g \sin \theta$$

$$\Rightarrow \frac{d^2(L\theta)}{dt^2} = -g \sin \theta \quad [x = L\theta]$$



$$\Rightarrow L \cdot \frac{d^2\theta}{dt^2} = -g \sin \theta$$

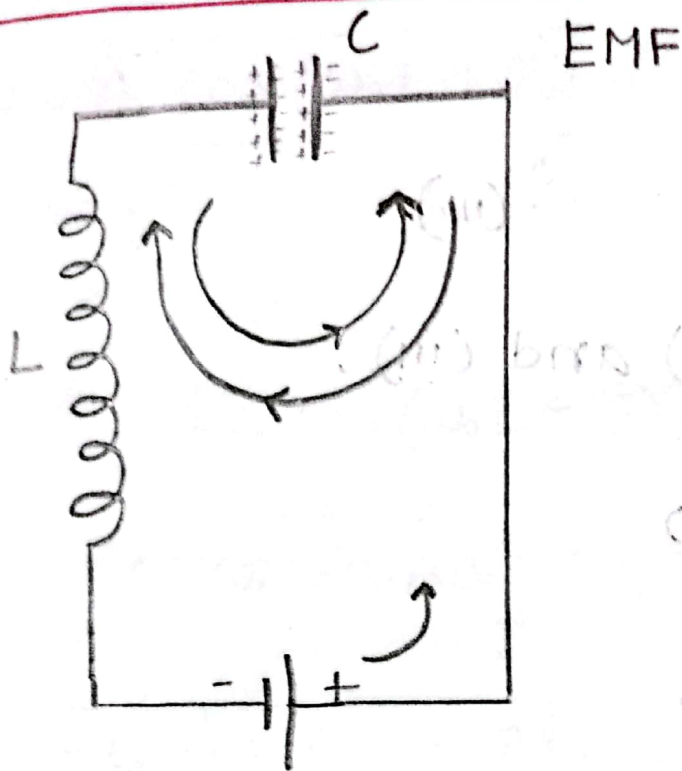
$$\Rightarrow L \cdot \frac{d^2\theta}{dt^2} = -g\theta \quad \begin{cases} \sin \theta < 1 \\ \sin \theta \approx \theta \end{cases}$$

$$\Rightarrow \frac{d^2\theta}{dt^2} = -\frac{g}{L} \theta$$

$$\Rightarrow \frac{d^2\theta}{dt^2} + \frac{g}{L} \theta = 0 \dots\dots (iii)$$

$$\Rightarrow \boxed{\frac{d^2\theta}{dt^2} + \omega_0^2 \theta = 0} \dots\dots (iv) \quad \left[\underline{\omega_0 = \sqrt{\frac{g}{L}}} \right]$$

Simple Harmonic Motion: LC Circuit (Tank/tuned circuit)



For LC circuit,

Voltage across Capacitor

$$V_c = \frac{Q}{C} \dots \dots \dots (i)$$

$Q = \text{charge}$
 $C = \text{Capacitance}$

Voltage across inductor

$$V_L = L \left(\frac{di}{dt} \right) \dots \dots \dots (ii)$$

From Kirchhoff's voltage law.

$$V_C = -V_L$$

$$\Rightarrow \boxed{V_C + V_L = 0} \dots \dots \dots (iii)$$

From eqn. (i), (ii) and (iii),

$$\frac{Q}{C} + L \left(\frac{di}{dt} \right) = 0$$

$$\Rightarrow \frac{Q}{LC} + \left(\frac{di}{dt} \right) = 0$$

$$\Rightarrow \frac{1}{LC} \left(\frac{dQ}{dt} \right) + \frac{d^2 i}{dt^2} = 0$$

$$\Rightarrow \left(\frac{1}{LC} \right) \dot{i} + \frac{d^2 i}{dt^2} = 0 \quad ; \quad \left[\dot{i} = \frac{dQ}{dt} \right]$$

$$\Rightarrow \frac{d^2 i}{dt^2} + \left(\frac{1}{LC} \right) \dot{i} = 0$$

$$\Rightarrow \boxed{\frac{d^2 i}{dt^2} + \omega_o^2 \dot{i} = 0}$$

$$(iv) \quad \left[\omega_o^2 = \frac{1}{LC} \right]$$

$$(v) \quad \left(\frac{1}{LC} \right) = \omega_o^2$$

Soln. of eqn. (iv)

$$\boxed{q = q_0 \cos(\omega_0 t + \varphi)} \dots \dots (i)$$

$$i = \frac{dq}{dt}$$

$$= \frac{d}{dt} [q_0 \cos(\omega_0 t + \varphi)]$$

$$= -q_0 \omega_0 \sin(\omega_0 t + \varphi)$$

$$\boxed{i = -i_0 \sin(\omega_0 t + \varphi)} \dots \dots (vi)$$

$$\text{frequency } f = \frac{\omega_0}{2\pi}$$

$$f = \frac{1}{2\pi} \sqrt{\frac{1}{LC}}$$

$$\boxed{f = \frac{1}{2\pi \sqrt{LC}}} \dots \dots (vii)$$

SHM for Complex plane

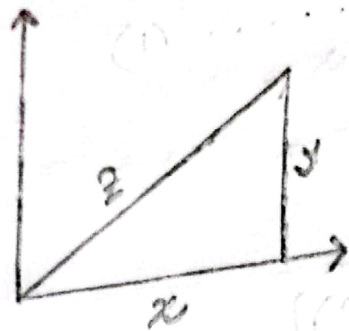


fig: SHM for complex plane.

For complex plane.

$$z = x + jy \dots (i)$$

$x = \text{real part}$
 $y = \text{imaginary part}$

Force, $f = -kz \dots (ii)$ $[F \propto -z]$ (not Hooke's Law)

From Newton's 2nd Law,

$$F = ma \dots (iii)$$

From eqn. (ii) and (iii),

$$ma = -kz$$

$$\Rightarrow m \left(\frac{dv}{dt} \right) = -kz$$

$$\Rightarrow m \cdot \frac{d}{dt} \left(\frac{dz}{dt} \right) = -kz$$

$$\Rightarrow m \cdot \left(\frac{d^2 z}{dt^2} \right) = -kz$$

$$\Rightarrow \frac{d^2 z}{dt^2} = -\frac{k}{m} z$$

$$\Rightarrow \frac{d^2 z}{dt^2} + \frac{k}{m} z = 0$$

$$\Rightarrow \boxed{\frac{d^2 z}{dt^2} + \omega_0^2 z = 0} \dots (iv) \quad [\omega_0 = \sqrt{k/m}]$$

Solⁿ of SHM in complex plane

$$\frac{d^2 z}{dt^2} + \omega_0^2 z = 0 \dots (iv)$$

$$\text{Let } z = B e^{pt} \dots (v)$$

$$\frac{dz}{dt} = \frac{d}{dt} (B e^{pt})$$

$$= B \cdot e^{pt} \cdot p$$

$$\frac{d^2 z}{dt^2} = \frac{d}{dt} \left(\frac{dz}{dt} \right)$$

$$= \frac{d}{dt} \{ B \cdot e^{pt} \cdot p \}$$

$$\therefore \frac{d^2 z}{dt^2} = B p^2 e^{pt} \dots (vi)$$

From eqⁿ (iv), (v) and (vi),

$$B p^2 e^{pt} + \omega_0^2 B e^{pt} = 0$$

$$\Rightarrow B e^{pt} (p^2 + \omega_0^2) = 0$$

$$\Rightarrow p^2 + \omega_0^2 = 0 ; [B e^{pt} \neq 0]$$

$$\Rightarrow p^2 = -\omega_0^2$$

$$\Rightarrow p = \sqrt{-\omega_0^2}$$

$$\Rightarrow p = \sqrt{-1} \cdot \sqrt{\omega_0^2}$$

$$\Rightarrow p = \pm j \omega_0 \dots \dots \dots (vii)$$

From eqⁿ (v) and (vii),

$$z = B e^{\pm j \omega_0 t} \dots \dots \dots (viii)$$

$$z = B_1 e^{j \omega_0 t} + B_2 e^{-j \omega_0 t} \dots \dots \dots (ix)$$

$$\therefore V = \frac{dz}{dt}$$

$$= \frac{d}{dt} \{ B_1 e^{j \omega_0 t} \} + \frac{d}{dt} \{ B_2 e^{-j \omega_0 t} \}$$

Graphical representation

$$(1) y = A \sin(\omega t + \pi/4)$$

$$y = 0$$

$$\Rightarrow A \sin(\omega t + \pi/4) = 0$$

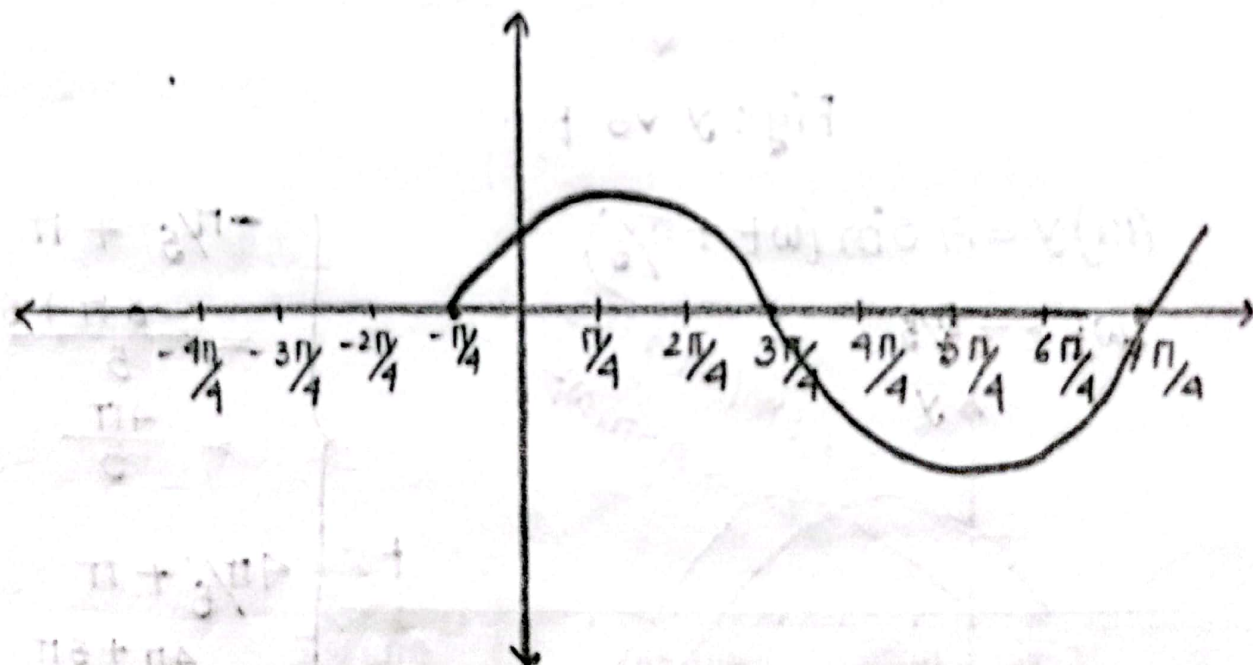
$$\Rightarrow \sin(\omega t + \pi/4) = 0$$

$$\Rightarrow \sin(\omega t + \pi/4) = \sin 0^\circ$$

$$\Rightarrow \omega t + \pi/4 = 0$$

$$\therefore \omega t = -\pi/4$$

$$\begin{cases} -\pi/4 + \pi = 3\pi/4 \\ 3\pi/4 + \pi = 7\pi/4 \end{cases}$$



$$(ii) y = A \sin(\omega t + \pi/7)$$

$y = 0$, then

$$\omega t = -\pi/7$$

$$\begin{cases} \pi/7 + \pi = 6\pi/7 \\ 6\pi/7 + \pi = 13\pi/7 \end{cases}$$

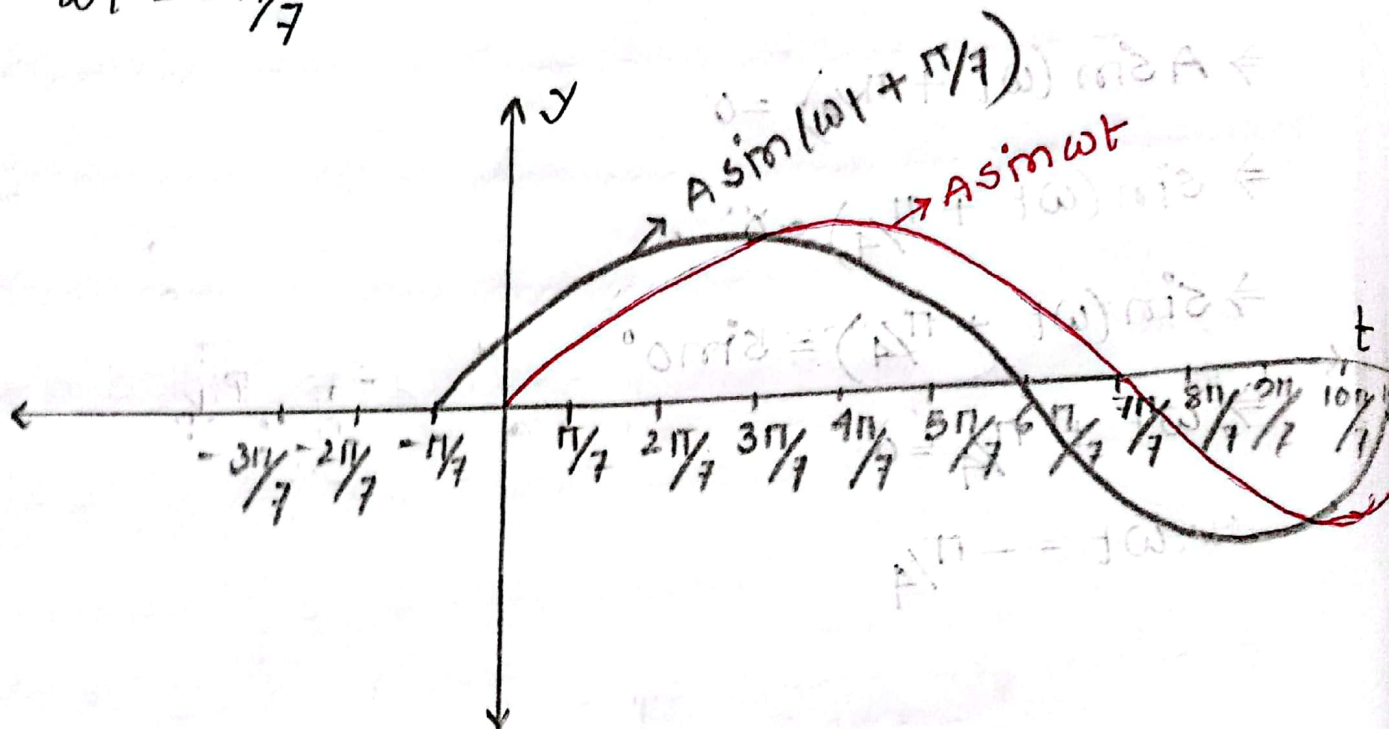
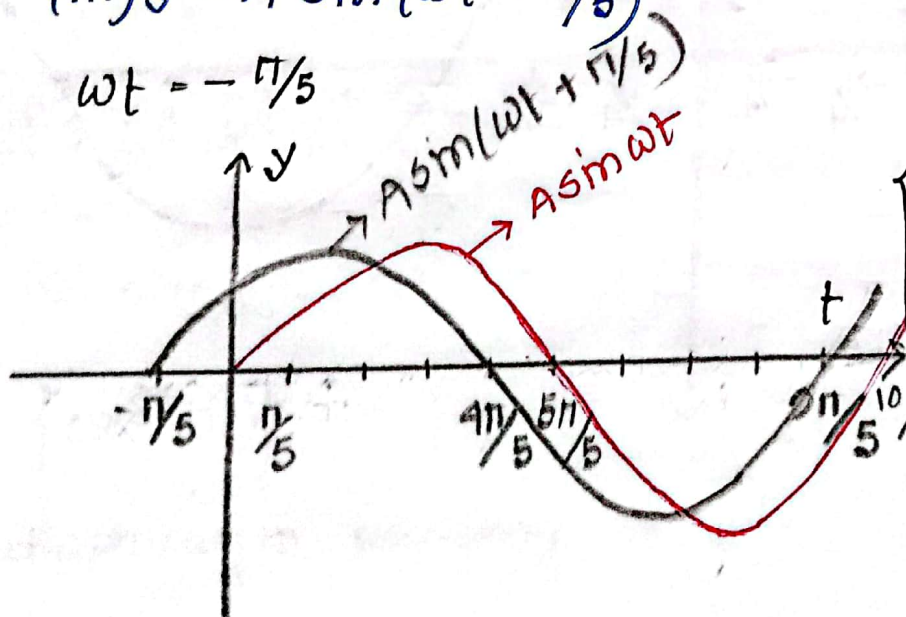


Fig: y vs t

$$(iii) y = A \sin(\omega t + \pi/5)$$

$$\omega t = -\pi/5$$

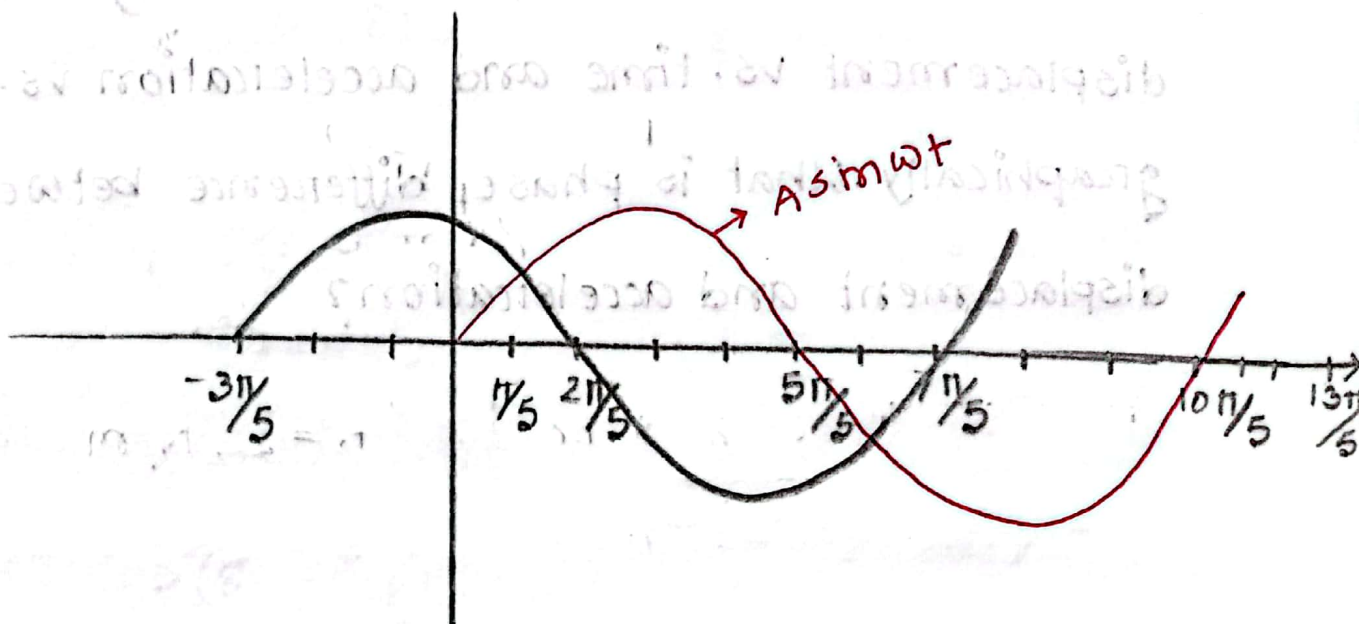


$$\begin{aligned} & -\pi/5 + \pi \\ &= \frac{-\pi + 5\pi}{5} \\ &= \frac{4\pi}{5} \end{aligned}$$

$$\begin{aligned} & 4\pi/5 + \pi \\ &= \frac{4\pi + 5\pi}{5} = 9\pi/5 \end{aligned}$$

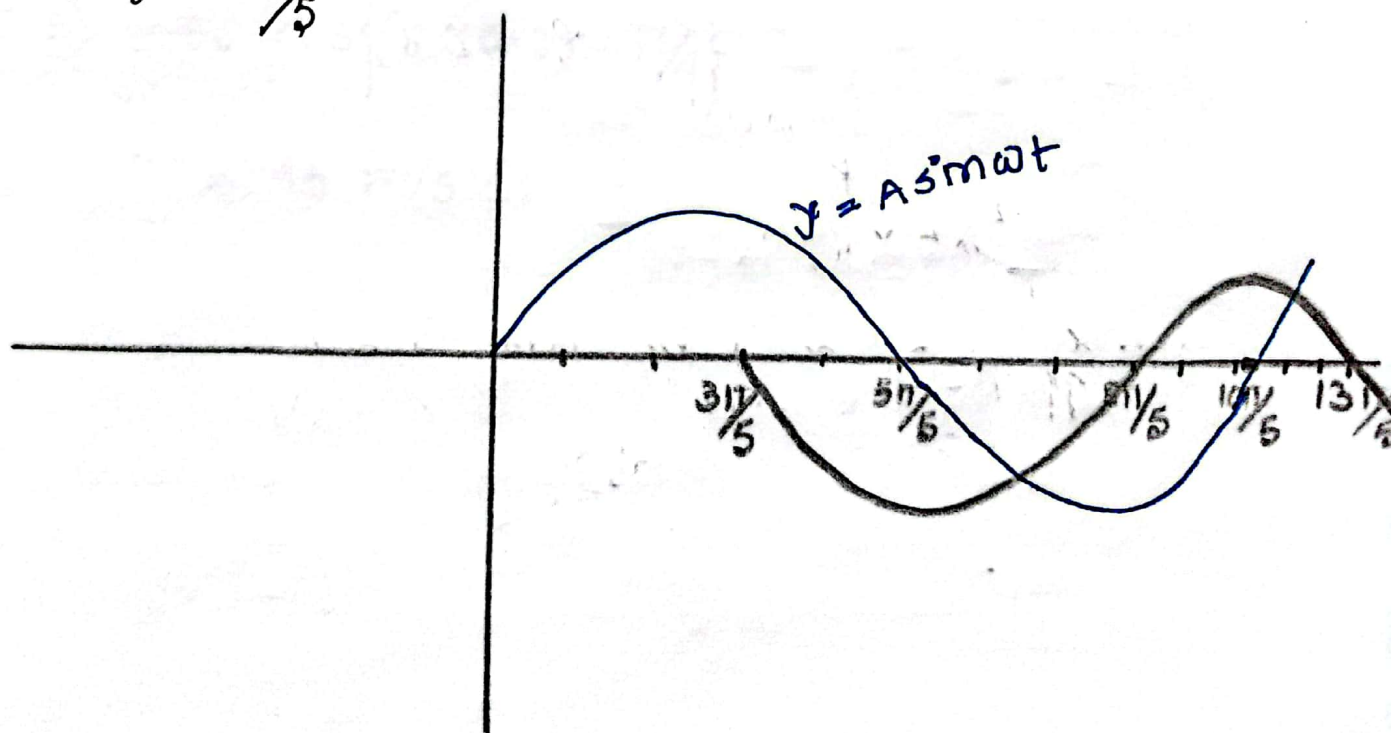
$$(iv) y = A \sin(\omega t + 3\pi/5)$$

$$\omega t = -3\pi/5$$



$$(v) y = -A \sin(\omega t - 3\pi/5)$$

$$\omega t = 3\pi/5$$



#Damped Harmonic Motion

Restoring force,

$$F = -kx + F_c \quad [F_c = \text{Frictional force}]$$

$$\Rightarrow F = -kx + (-bv)$$

$$\Rightarrow F = -kx - bv \dots (i)$$

$$\begin{cases} b = \text{Frictional constant} \\ v = \text{velocity} \\ F_c = -bv \end{cases}$$

From Newton's 2nd Law,

$$F = ma \dots (ii)$$

From eqⁿ (i) and (ii);

$$ma = -kx - bv$$

$$\Rightarrow m \left(\frac{d^2x}{dt^2} \right) = -kx - b \left(\frac{dx}{dt} \right)$$

$$\Rightarrow \frac{d^2x}{dt^2} = -\frac{k}{m}x - \frac{b}{m} \left(\frac{dx}{dt} \right)$$

$$\Rightarrow \frac{d^2x}{dt^2} + \frac{k}{m}x + \frac{b}{m} \left(\frac{dx}{dt} \right) = 0$$

$$\Rightarrow \boxed{\frac{d^2x}{dt^2} + \left(\frac{b}{m} \right) \frac{dx}{dt} + \left(\frac{k}{m} \right) x = 0} \dots (iii)$$

$$\Rightarrow \boxed{\frac{d^2x}{dt^2} + \gamma \cdot \frac{dx}{dt} + \omega_0^2 x = 0} \dots \dots (iv)$$

$[\gamma = \text{damping constant}]$

Solⁿ for damped harmonic motion

Let the trial solution for equation (iv),

$$x = B e^{pt} \dots \dots \dots (v)$$

Here, $\boxed{p = -\frac{\gamma}{2} \pm \sqrt{\left(\frac{\gamma^2}{4} - \omega_0^2\right)}} \dots \dots \dots (vi)$

Case - 01

$$A = A_0 e^{-\gamma/2 t}$$

When $\omega_0^2 > \frac{\gamma^2}{4}$ ✓ (xi) $\tau = 1/\gamma$

$$E = E_0 e^{-2\alpha t}$$

From eqⁿ (vi),

$$p = -\frac{\gamma}{2} \pm \sqrt{(\omega_0^2 - \frac{\gamma^2}{4})}$$

$$= -\frac{\gamma}{2} \pm \sqrt{-\omega_1^2}; \quad [\omega_1^2 = (\omega_0^2 - \frac{\gamma^2}{4})]$$

$$\therefore p = -\frac{\gamma}{2} \pm j\omega_1 \dots \dots \dots (vii)$$

From eqn (v) and (vii),

$$x = B e^{(-\gamma/2 \pm j\omega_1)t}$$

$$= B_1 e^{(-\gamma/2 + j\omega_1)t} + B_2 e^{(-\gamma/2 - j\omega_1)t}$$

$$x = B_1 e^{-\gamma/2 + j\{\sqrt{\omega_0^2 + \gamma^2/4}\}t} + B_2 e^{-\gamma/2 - j\{\sqrt{\omega_0^2 + \gamma^2/4}\}t} \dots \dots (viii)$$

Case-02

When $\omega_0^2 < \gamma^2/4$

From eqn (vi),

$$p = -\gamma/2 \pm \lambda \dots \dots (ix) \quad \left[\lambda = \sqrt{\gamma^2/4 - \omega_0^2} \right]$$

From eqn (v) and (ix),

$$x = B e^{(-\gamma/2 \pm \lambda)t}$$

$$= B_1 e^{(-\gamma/2 + \lambda)t} + B_2 e^{(-\gamma/2 - \lambda)t}$$

$$x = B_1 e^{\{-\gamma/2 + \sqrt{\gamma^2/4 - \omega_0^2}\}t} + B_2 e^{\{-\gamma/2 - \sqrt{\gamma^2/4 - \omega_0^2}\}t} \dots \dots (x)$$

Case - 03

When $\omega_0^2 = \frac{\gamma^2}{4}$.

From eqⁿ (vi),

$$P = -\frac{\gamma}{2} \dots \dots \dots (xi)$$

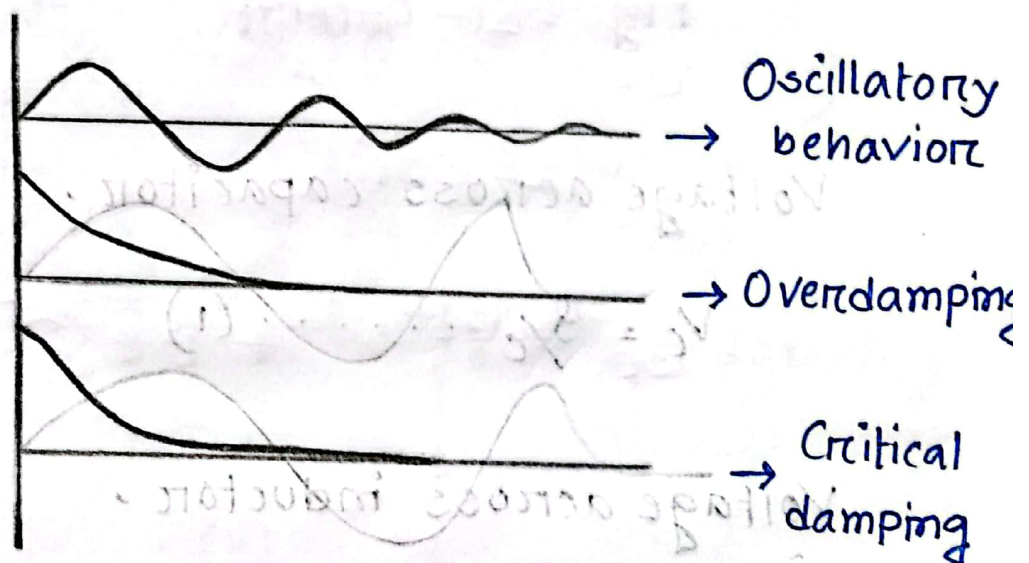
From eqⁿ (v) and (xi),

$$x = B e^{(-\gamma/2)t} \dots \dots \dots (xii)$$

(i) $\omega_0^2 > \frac{\gamma^2}{4}$

(ii) $\omega_0^2 < \frac{\gamma^2}{4}$

(iii) $\omega_0^2 = \frac{\gamma^2}{4}$



Damped Oscillation: RLC Circuit

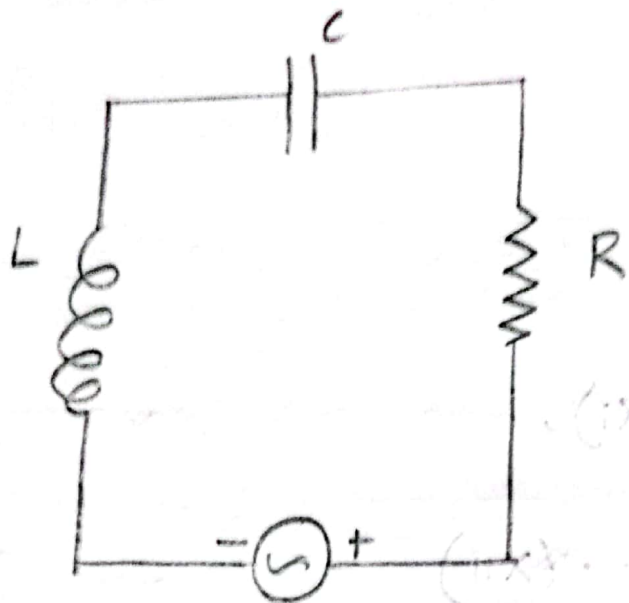


Fig: LRC circuit

Voltage across capacitor,

$$V_C = q/c \dots \dots \dots (i)$$

Voltage across inductor,

$$V_L = L (di/dt) \dots \dots \dots (ii)$$

Voltage across resistor,

$$V_R = iR \dots \dots \dots (iii)$$

Applying Kirchhoff's voltage Law,

$$V_L + V_C + V_R = 0$$

$$\Rightarrow L \left(\frac{di}{dt} \right) + \frac{Q}{C} + iR = 0$$

$$\Rightarrow \frac{di}{dt} + \left(\frac{1}{LC} \right) Q + \left(\frac{R}{L} \right) i = 0$$

$$\Rightarrow \frac{d}{dt} \left(\frac{dQ}{dt} \right) + \left(\frac{1}{LC} \right) Q + \left(\frac{R}{L} \right) \frac{dQ}{dt} = 0$$

$$\Rightarrow \frac{d^2 Q}{dt^2} + \left(\frac{R}{L} \right) \frac{dQ}{dt} + \left(\frac{1}{LC} \right) Q = 0 \quad \dots \dots (iii)$$

$$\Rightarrow \boxed{\frac{d^2 Q}{dt^2} + \gamma \frac{dQ}{dt} + \omega_0^2 Q = 0} \quad \dots \dots (iv)$$

$\gamma = R/L$; damping constant

let,

$$Q = B e^{Pt} \quad \dots \dots (v)$$

Here,

$$P = -\gamma/2 \pm \sqrt{(\gamma^2/4 - \omega_0^2)} \quad \dots \dots (vi)$$

case - 02

$$\omega_0^2 < \gamma^2/4$$

$$\begin{aligned} g &= B_1 e^{\{-\gamma/2 + \sqrt{(\gamma^2/4 - \omega_0^2)}\} t} + B_2 e^{\{-\gamma/2 - \sqrt{(\gamma^2/4 - \omega_0^2)}\} t} \\ &= B_1 e^{-\{R/2L + \sqrt{(R^2/4L^2 - 1/LC)}\} t} + B_2 e^{-\{R/2L - \sqrt{(R^2/4L^2 - 1/LC)}\} t} \end{aligned}$$

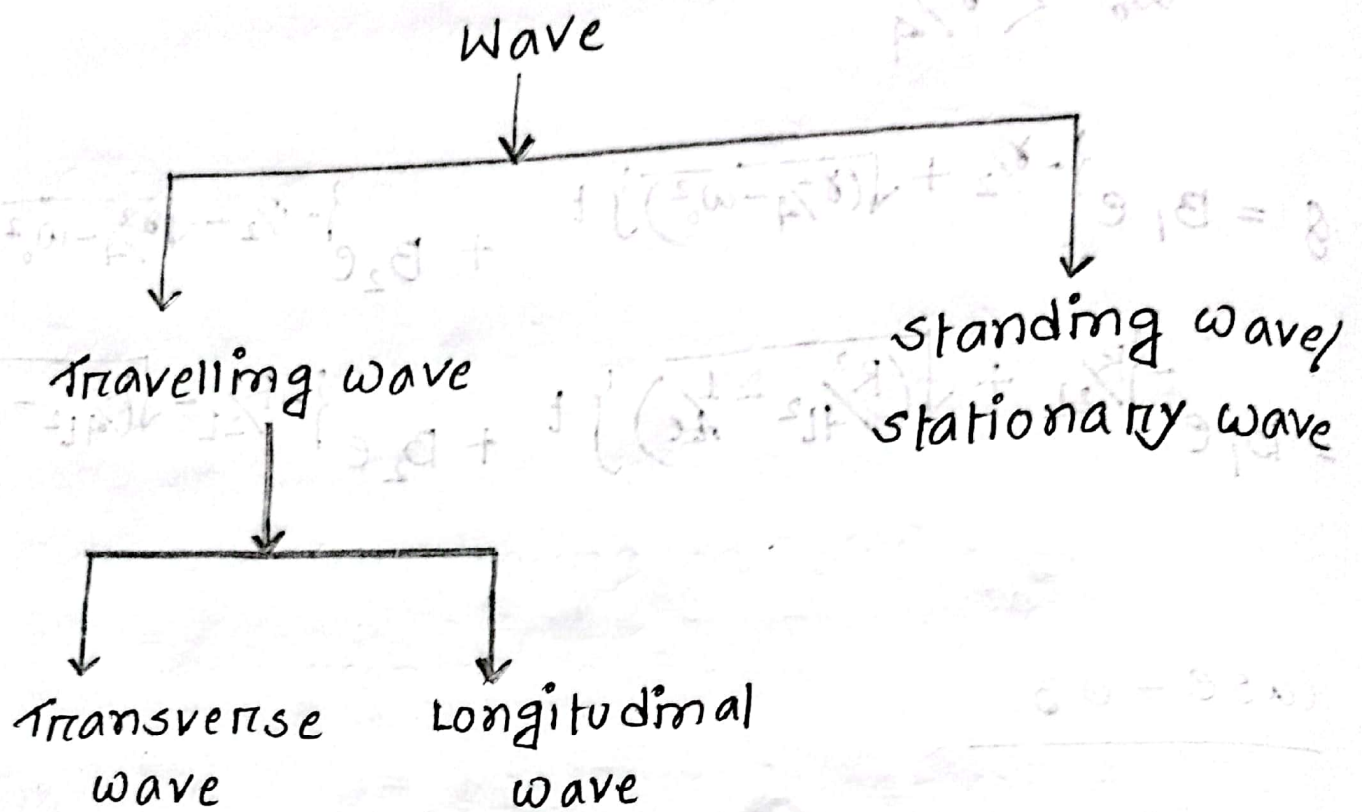
case - 03

$$\omega_0^2 = \gamma^2/4$$

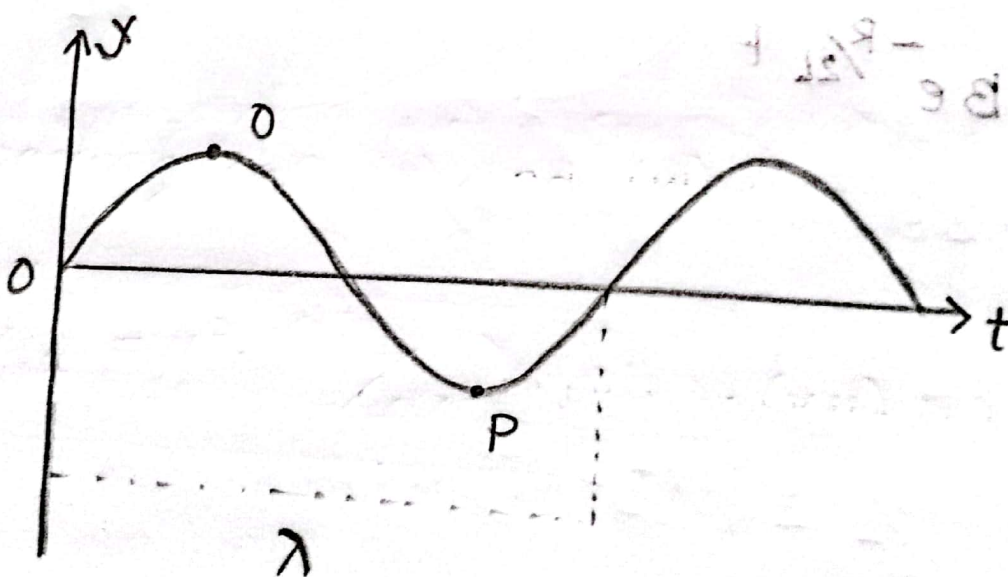
$$g = B e^{-\gamma/2 t}$$

$$\therefore g = B e^{-R/2L t}$$

Wave



Wave equation for travelling wave



For particle O, displacement is -

$$y_1 = A \sin(\omega t + \phi) \dots \dots \dots (i)$$

For particle P,

displacement is,

$$y_2 = A \sin(\omega t - \phi) \dots \dots \dots (ii)$$

$$= A \sin\left(\omega t - \frac{2\pi x}{\lambda}\right) \left[\phi = \frac{2\pi x}{\lambda}\right]$$

$$= A \sin\left(2\pi/T \cdot t - \frac{2\pi x}{\lambda}\right)$$

$$= A \sin\left(\frac{2\pi vt}{\lambda} - \frac{2\pi x}{\lambda}\right)$$

$$\boxed{y_2 = A \sin \frac{2\pi}{\lambda} (vt - x)} \dots \dots \dots (iii)$$

$$\begin{cases} s = vt \\ \lambda = vT \\ T = \lambda/v \\ 1/T = v/\lambda \end{cases}$$

From eqn (i),

$$\boxed{y_1 = A \sin \frac{2\pi}{\lambda} (vt + x)} \dots \dots \dots (iv)$$

Now, differentiating eqn (iv) with respect to x ,

$$\frac{dy_1}{dx} = A \cos \frac{2\pi}{\lambda} (vt + x) \left(\frac{2\pi}{\lambda} \right)$$

$$\frac{d^2 y_1}{dx^2} = -A \left(\frac{2\pi}{\lambda} \right)^2 \sin \frac{2\pi}{\lambda} (vt + x) \dots \dots (v)$$

Again, differentiating eqn (iv) with respect to t ,

$$\frac{dy_1}{dt} = A \cos \frac{2\pi}{\lambda} (vt + x) \left(\frac{2\pi v}{\lambda} \right)$$

$$\frac{d^2 y_1}{dt^2} = -A \cos \left(\frac{2\pi}{\lambda} \right)^2 v^2 \sin \frac{2\pi}{\lambda} (vt + x) \dots \dots (vi)$$

Comparing eqn (v) and (vi),

$$\boxed{\frac{d^2 y_1}{dt^2} = v^2 \cdot \left(\frac{d^2 y_1}{dx^2} \right)} \dots \dots (vii)$$

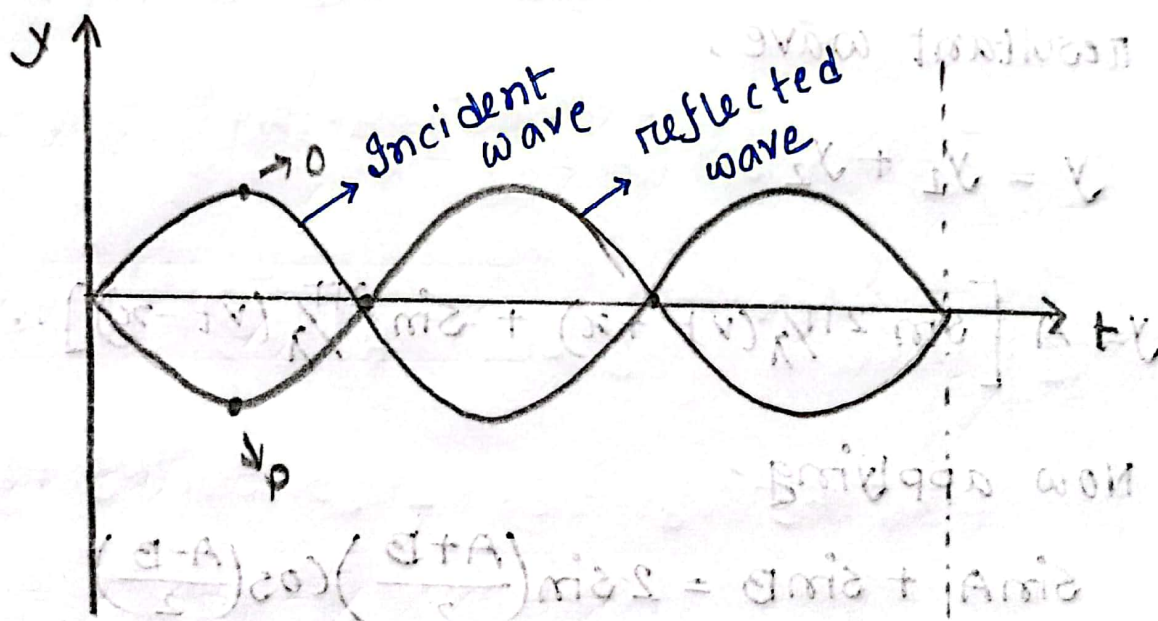
From eqn (v),

$$\boxed{\frac{d^2 y_1}{dx^2} = -\left(\frac{2\pi}{\lambda}\right)^2 y_1 \dots \dots (vii)}$$

From eqn (vi),

$$\boxed{\frac{d^2 y_1}{dt^2} = -\left(\frac{2\pi}{\lambda}\right)^2 v^2 y_1 \dots \dots (viii)}$$

Wave equation for Standing wave



For Particle O,

displacement is,

$$y_1 = A \sin (\omega t + \varphi) \dots \dots (i)$$

For Particle P,

displacement is -

$$y_2 = A \sin(\omega t - \phi) \dots \dots \dots (ii)$$

From eqn (i) and (ii),

$$y_1 = A \sin \frac{2\pi}{\lambda} (vt + x) \dots \dots \dots (iii)$$

$$y_2 = A \sin \frac{2\pi}{\lambda} (vt - x) \dots \dots \dots (iv)$$

By adding eqn (iii) and (iv) we get the resultant wave,

$$y = y_1 + y_2$$

$$y = A \left[\sin \frac{2\pi}{\lambda} (vt + x) + \sin \frac{2\pi}{\lambda} (vt - x) \right] \dots \dots (v)$$

Now applying -

$$\sin A + \sin B = 2 \sin \left(\frac{A+B}{2} \right) \cos \left(\frac{A-B}{2} \right)$$

From eqn (v),

$$y = A \left[2 \sin \left\{ \frac{\frac{2\pi}{\lambda} (vt + x) + \frac{2\pi}{\lambda} (vt - x)}{2} \right\} \cos \left\{ \frac{\frac{2\pi}{\lambda} (vt + x) - \frac{2\pi}{\lambda} (vt - x)}{2} \right\} \right]$$

$$\Rightarrow y = A \left[2 \sin\left(\frac{2\pi}{\lambda} vt\right) \cos\left(\frac{2\pi}{\lambda} x\right) \right]$$

$$\therefore y = 2A \sin\left(\frac{2\pi vt}{\lambda}\right) \cos\left(\frac{2\pi x}{\lambda}\right) \dots \dots (vi)$$

$$\therefore y = 2A \sin(\omega t) \cos(kx) \dots \dots (vii)$$

• Suppose the displacement of a DHM is expressed as $x = Ae^{-\alpha t} \cos(\omega_d t + \delta)$. Show that total energy of DHM is $E = E_0 e^{-2\alpha t}$.

$$x = Ae^{-\alpha t} \cos(\omega_d t + \delta) \dots \dots (i)$$

$$\therefore v = \frac{dx}{dt} = A \left\{ -e^{-\alpha t} \sin(\omega_d t + \delta) \cdot \omega_d + \cos(\omega_d t + \delta) \cdot (-\alpha \cdot e^{-\alpha t}) \right\}$$

$$v = -A \omega_d e^{-\alpha t} \sin(\omega_d t + \delta) - A \alpha e^{-\alpha t} \cos(\omega_d t + \delta)$$

$$\therefore v^2 = \left\{ -A \omega_d e^{-\alpha t} \sin(\omega_d t + \delta) - A \alpha e^{-\alpha t} \cos(\omega_d t + \delta) \right\}^2$$

$$= A^2 \omega_d^2 e^{-2\alpha t} \sin^2(\omega_d t + \delta) + A^2 \alpha^2 e^{-2\alpha t} \cos^2(\omega_d t + \delta) + 2 A^2 \omega_d \alpha e^{-2\alpha t} \sin(\omega_d t + \delta) \cos(\omega_d t + \delta)$$

From eqn (1)

$$x = A e^{-\alpha t} \cos(\omega_d t + \delta)$$

$$\Rightarrow x^2 = A^2 e^{-2\alpha t} \cos^2(\omega_d t + \delta)$$

$$\begin{cases} \omega_d^2 = (\omega_0^2 - \gamma^2/4) \\ \Rightarrow \omega_d^2 = \omega_0^2 - \alpha^2 \\ \Rightarrow \omega_0^2 = \omega_d^2 + \alpha^2 \end{cases}$$

We know,

$$E = \frac{1}{2} m v^2 + \frac{1}{2} k x^2$$

$$= \frac{1}{2} m \left[A^2 \omega_d^2 e^{-2\alpha t} \sin^2(\omega_d t + \delta) + A^2 \alpha^2 e^{-2\alpha t} \cos^2(\omega_d t + \delta) \right. \\ \left. + 2 A^2 \omega_d \alpha e^{-2\alpha t} \sin(\omega_d t + \delta) \cos(\omega_d t + \delta) \right]$$

$$+ \frac{1}{2} k \left\{ A^2 e^{-2\alpha t} \cos^2(\omega_d t + \delta) \right\} - \left[\omega_0^2 = k/m \right]$$

$$= \frac{1}{2} m A^2 e^{-2\alpha t} \left[\omega_d^2 \sin^2(\omega_d t + \delta) + \alpha^2 \cos^2(\omega_d t + \delta) \right. \\ \left. + \omega_0^2 \cos^2(\omega_d t + \delta) + 2 \omega_d \alpha \sin(\omega_d t + \delta) \cos(\omega_d t + \delta) \right]$$

$$= \frac{1}{2} m A^2 e^{-2\alpha t} \left[\omega_0^2 \sin^2(\omega_d t + \delta) - \alpha^2 \sin^2(\omega_d t + \delta) \right. \\ \left. + \alpha^2 \cos^2(\omega_d t + \delta) + \omega_0^2 \cos^2(\omega_d t + \delta) + 2 \omega_d \alpha \sin(\omega_d t + \delta) \cos(\omega_d t + \delta) \right]$$

$$= \frac{1}{2} m A^2 e^{-2\alpha t} \left[\omega_d^2 \sin^2(\omega_d t + \delta) + \alpha^2 \cos^2(\omega_d t + \delta) \right. \\ \left. + \omega_d^2 \cos^2(\omega_d t + \delta) + \alpha^2 \cos^2(\omega_d t + \delta) \right. \\ \left. + 2 \omega_d \alpha \sin(\omega_d t + \delta) \cos(\omega_d t + \delta) \right]$$

$$= \frac{1}{2} m A^2 e^{-2\alpha t} \left[1 + \underbrace{2 \alpha^2 \cos^2(\omega_d t + \delta)}_{\text{negligible}} + 2 \alpha \omega_d \sin(\omega_d t + \delta) \cos(\omega_d t + \delta) \right]$$

$$= \frac{1}{2} m A^2 e^{-2\alpha t} \left[1 + 2 \omega_d \alpha \sin(\omega_d t + \delta) \cos(\omega_d t + \delta) \right]$$

$$= \frac{1}{2} m A^2 e^{-2\alpha t} \left[1 + \omega_d \alpha \sin 2(\omega_d t + \delta) \right]$$

$$= \frac{1}{2} m A^2 e^{-2\alpha t} [1 + 1]$$

$$= m A^2 e^{-2\alpha t}$$

$$= E_0 e^{-2\alpha t} \quad [E_0 = m A^2]$$

$$\therefore E = E_0 e^{-2\alpha t}$$

(Proved)